Identifying Hidden Market Regimes from Financial Time-Series Data Using Probabilistic State-Space Models

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***Abstract*—Financial markets exhibit non-stationary and regime-dependent behaviour driven by latent factors that are not directly observable. This work extends the application of probabilistic state-space models, particularly Hidden Markov Models (HMMs), to identify such hidden regimes in financial time-series data. The study formalises the role of latent random variables, Markovian dynamics, and regime-switching models, and presents a unified treatment of inference and learning using the Forward-Backward procedure, Viterbi decoding, and the Expectation-Maximisation (EM) framework. Gaussian Mixture Models (GMMs) are also examined as a precursor for modelling emission distributions. The proposed methodology leverages the Baum–Welch algorithm for parameter estimation and enables robust identification of underlying market regimes.**

***Index Terms*—Hidden Markov Models, Latent Variables, Financial Time-Series, Baum–Welch Algorithm, Regime Detection**

I. Introduction

Financial time-series exhibit stochastic dynamics with statistical properties that evolve over time due to latent economic and behavioural factors. Empirical evidence shows that asset returns display volatility clustering, heavy tails, and structural breaks, which cannot be captured by static models such as Gaussian i.i.d. processes [1]. This motivates the use of models that incorporate time-varying structure and hidden dynamics.

Latent variable models address this by introducing unobserved variables governing the data generation process. In particular, regime-switching models assume that financial time-series evolve across a finite set of hidden states, each representing a distinct market regime.

Hidden Markov Models (HMMs) extend Markov chains by combining latent state dynamics with state-dependent observation distributions [2], [6]. This enables modelling of both temporal dependencies and hidden structure. Building on prior work [3], this study applies HMM-based methods for regime identification, including inference and parameter estimation techniques.

II. Literature Review

Markov chains form the basis of sequential probabilistic models by assuming that the current state depends only on the previous state, leading to the factorisation

$$p(z_{1:T}) = p(z_1) \prod_{t=2}^{T} p(z_t|z_{t-1}), \tag{1}$$

which enables efficient inference [2].

Hidden Markov Models (HMMs) extend this framework by introducing latent states $z_t \in \{1, \dots, K\}$ that generate observations x_t, with joint distribution

$$p(z_{1:T}, x_{1:T}) = p(z_1) \prod_{t=2}^{T} p(z_t|z_{t-1}) \prod_{t=1}^{T} p(x_t|z_t), \tag{2}$$

allowing compact modelling of temporal dependencies [6].

In financial applications, HMMs effectively capture regime-switching behaviour, including volatility clustering and structural changes [3], [4]. Hybrid approaches combining HMMs with forecasting methods have also been explored [5].

Inference in HMMs is performed using dynamic programming. The Forward-Backward algorithm computes posterior probabilities using

$$\alpha_t(j) = p(x_{1:t}, z_t = j), \quad \beta_t(j) = p(x_{t+1:T}|z_t = j), \tag{3}$$

while the Viterbi algorithm determines the most probable state sequence

$$z^*_{1:T} = \arg\max_{z_{1:T}} p(z_{1:T}|x_{1:T}). \tag{4}$$

Parameter estimation is carried out using the Expectation-Maximisation (EM) framework via the Baum–Welch algorithm [6]. Gaussian Mixture Models (GMMs),

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x|\mu_k, \Sigma_k), \tag{5}$$

are commonly used for initialising emission parameters [2].

Recent studies further explore reinforcement learning-based regime adaptation and Bayesian changepoint detection methods [7].

III. Proposed Methodology

The objective of this work is to identify hidden market regimes from financial time-series data by leveraging the probabilistic structure of Hidden Markov Models (HMMs). Let $\{x_t\}_{t=1}^T$ denote the observed sequence of daily log-returns of a financial asset or index (e.g., S&P 500), and $\{z_t\}_{t=1}^T$ denote the corresponding latent regime sequence. Each hidden state $z_t \in \{1, \ldots, K\}$ represents an underlying market condition such as bull, bear, or high-volatility regimes.

The modelling assumption is that financial returns are conditionally Gaussian given the regime, while regime transitions follow a first-order Markov process. The transition probabilities are defined as

$$A_{ij} = p(z_t = j | z_{t-1} = i), \tag{6}$$

which capture the persistence and switching behaviour of market regimes. The emission distribution for each regime is given by

$$p(x_t | z_t = j) = \mathcal{N}(x_t | \mu_j, \Sigma_j), \tag{7}$$

where μ_j and Σ_j represent the mean return and volatility associated with regime j. This formulation allows different regimes to exhibit distinct statistical characteristics, such as low mean–low volatility (stable markets) or low mean–high volatility (crisis periods).

To initialise the emission parameters, Gaussian Mixture Models (GMMs) are fitted to the return distribution, providing a data-driven approximation of regime clusters. These initial estimates are then refined using the Baum–Welch algorithm, which iteratively maximises the likelihood of observed returns. During the E-step, the Forward-Backward algorithm computes the posterior probabilities $p(z_t | x_{1:T})$, representing the likelihood of each regime at every time step. In the M-step, these probabilities are used to update transition and emission parameters through weighted maximum likelihood estimation.

Once the model parameters converge, regime inference is performed in two complementary ways. First, the smoothed posterior probabilities obtained from the Forward-Backward algorithm provide a probabilistic view of regime membership over time, enabling analysis of gradual transitions and uncertainty in regime classification. Second, the Viterbi algorithm is applied to compute the most probable sequence of hidden states,

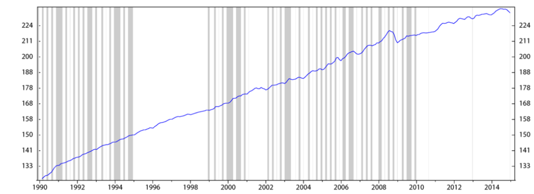
$$z_{1:T}^* = \arg\max_{z_{1:T}} p(z_{1:T} | x_{1:T}), \tag{8}$$

which yields a discrete segmentation of the time-series into distinct regimes.

The identified regimes are then interpreted in financial terms by analysing regime-specific statistics such as mean returns, volatility, and duration.

Limitations of HMM: Hidden Markov Models have certain limitations in financial applications. The Gaussian assumption may not capture heavy tails in returns, while the first-order Markov property ignores long-term dependencies. The model is also sensitive to the choice of hidden states and initialisation, and its assumption of abrupt regime switches may not reflect gradual market transitions.

IV. Results



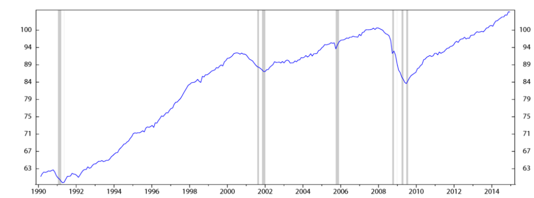
Regime Parameters (2014-12-31)

Mu		Sigma	
Inflation (Unshaded)	Deflation (Shaded)	Inflation (Unshaded)	Deflation (Shaded)
2.80	7.28	0.93	3.55

Data Statistics

Mean		Variance	
Inflation (Unshaded)	Deflation (Shaded)	Inflation (Unshaded)	Deflation (Shaded)
2.37	2.89	0.69	2.26

Fig. 1. Regimes of inflation, CPI; monthly data from January 1990 to December 2014 (log scale).



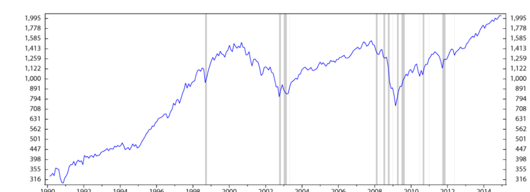
Regime Parameters (2014-12-31)

Mu		Sigma	
Growth (Unshaded)	Recession (Shaded)	Growth (Unshaded)	Recession (Shaded)
0.31	-0.47	0.51	1.27

Data Statistics

Mean		Variance	
Growth (Unshaded)	Recession (Shaded)	Growth (Unshaded)	Recession (Shaded)
0.21	-0.57	0.24	1.82

Fig. 2. Regimes of the industrial production index, INDPRO; monthly data from January 1990 to December 2014 (log scale).



Regime Parameters (2014-12-31)

Mu		Sigma	
Bull (Unshaded)	Bear (Shaded)	Bull (Unshaded)	Bear (Shaded)
0.57	0.97	0.51	1.27

Data Statistics

Mean		Variance	
Bull (Unshaded)	Bear (Shaded)	Bull (Unshaded)	Bear (Shaded)
1.02	-0.76	12.30	59.48

Fig. 3. Regimes of the stock market index, S&P 500; monthly data from January 1990 to December 2014 (log scale).

References

[1] J. D. Hamilton, "A new approach to the economic analysis of nonstationary time series and the business cycle," *Econometrica*, vol. 57, no. 2, pp. 357–384, 1989.

[2] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.

[3] Y. Yuan and G. Mitra, "Market regime identification using hidden Markov models," SSRN, 2019.

[4] P. Nystrup, H. Madsen, and E. Lindstrom, "Regime-switching factor investing with hidden Markov models," *J. Risk Financ. Manag.*, 2020.

[5] W. Thongkam and T. Kasetkasem, "Implementation of hidden Markov model to predict financial market regime," *IEEE ICSEC*, 2021.

[6] K. P. Murphy, *Machine Learning: A Probabilistic Perspective*. MIT Press, 2012.

[7] X. Chen, "HMM-based market regime detection with RL for portfolio management," *IDS*, 2025.

[8] N. Nguyen and D. Nguyen, "Hidden Markov Model for Stock Selection," *Risks*, vol. 3, no. 4, pp. 455–473, Oct. 2015.

[9] J. McGreevy, "Detecting multivariate market regimes via clustering algorithms," Imperial College London, 2023.